Renewable Energy: A Cheaper & More Stable Future for Hawai‘i

Portfolio edition: minor spelling, formatting, and privacy cleanup only. Content reflects the original date and context.

In the past few weeks, Hawaiian Electric has commissioned two major solar and paired battery projects: K■pono Solar on O‘ahu and AES K■ihelani on Maui. These projects, part of the Hawaiian Electric Stage 1 & 2 initiatives, have overcome significant challenges from the COVID-19 pandemic and supply chain disruptions to expand the state's clean energy portfolio and lower the state’s carbon footprint. However, an often-overlooked benefit is the cost stability and savings provided by this new energy source. Solar energy is substantially cheaper than fossil fuels have been in recent years. Additionally, the price of renewable energy remains constant for the project's lifetime, protecting the state from global oil price fluctuations that we would otherwise be using.

Stage 1 & 2 Projects

The K■pono Solar & AES K■ihelani projects are accompanied by other Stage 1 & 2 RFPs from Hawaiian Electric’s Integrated Grid Plan, part of a large-scale effort to adopt more renewable energy in the state. Since 2022, Hawai‘i has commissioned six new utility-scale solar and battery projects within the Hawaiian Electric service territories, with four more projects set to come online throughout 2024 and 2025. These clean energy initiatives will add a total of 338.5 MW of variable generation capacity, supported by 1481 MWh of battery storage. Combined, these projects have the potential to produce 895 GWh of electricity annually, representing over 10% of the total electricity sales in 2023.

Significant Cost Savings

[See Understanding Electricity Pricing for a primer on how electricity pricing is determined.]

In addition to aiding Hawai‘i's progress toward its 2045 mandate of 100% renewable energy, the energy purchased from new renewable energy facilities is substantially cheaper and offers a stable price for the next 20-25 years. The cost of energy from the new projects is 11.08 ¢/kWh, less than half of the average cost for the state’s utilities to generate their own energy at 20.66 ¢/kWh. When all projects are fully operational in 2025, the state will save nearly $86 million by buying 895 GWh of electricity from these renewable sources rather than on generation fueled by foreign oil imports, based on 2023 fuel and avoided energy costs. This could save the average household $5.70 per month assuming an average usage of 500 kWh. Moreover, if households could buy 100% of their energy from these new facilities, they would save nearly $48 a month!

Switching to renewable energy not only supports Hawai‘i's renewable energy mandate but also offers significant financial benefits. By locking in lower prices through new solar and battery projects, Hawai‘i can achieve substantial savings and provide more stable energy costs for its residents.

Upcoming Projects

While the addition of the Stage 1 & 2 Projects is substantial, the continued addition of diverse renewable energy sources is essential for achieving the State’s energy goals. Hawaiian Electric’s Stage 3 RFP Projects have recently been selected and contracts are being negotiated and presented to the commission. While prices are not yet available, you can review the proposed projects here.

Understanding Electricity Pricing

As a quick overview of how electricity pricing works, the utility buys, distributes, and sells energy on their electricity network. The costs are combined and spread out over the amount of electricity customers use according to a set rate (¢/kWh) governed by the Public Utilities Commission.

Utility Energy Purchases: The utility must either buy electricity from Independent Power Producers, distributed energy sources, or produce its own.

Competitive Bidding for Large-Scale Projects (IPPs): For large-scale energy purchases, utilities allow companies to competitively bid on projects to keep costs low for customers. This results in a power purchase agreement (PPA) that usually lasts 20-25 years, with prices set at a constant rate.

Tariffs for Small Energy Sales: Households can sell excess energy from their own solar systems to the utility at a pre-determined tariff.

Utility-Generated Energy: Utilities also generate their own electricity. In Hawai‘i this is primarily done using imported oil, which is both volatile and costly. This cost for a utility to generate its own energy is known as the avoided cost.

Non-Energy Purchases (omitted in these figures):

Grid Balancing: Maintaining a constant balance between supply and demand on the state’s six separate grids.

Operations and Maintenance: Maintaining operations and upkeep.

Miscellaneous Projects: Various safety projects and planning for changes to the grid.